RR-G-691B
January 15, 1973
SUPERSEDING
Fed. Spec. RR-G-691A
May 1, 1963

# FEDERAL SPECIFICATION

## GRIP, CABLE JAW

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

### 1. SCOPE AND CLASSIFICATION

1.1 This specification covers jaw-type grips used for holding taut and pulling wires, cables, and wire rope and are not intended for temporary fittings. Wedge grips which may be used as permanent fittings are covered in Federal Specification RR-G-700.

1.1.1 Federal specification coverage. Federal specifications do not include all varieties of the commodity indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Types and styles. Grips shall be of the following types and styles as specified (see 6.1):

Type I - Parallel jaw.
  Style A - Smooth jaws.

Type II - Eccentric jaw.

### 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specification:

PPP-P-40 - Packaging and Packing of Hand Tools.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.



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Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Illustrations. The illustrations shown herein are for convenience of identification and are not intended to preclude the purchase of grips which are otherwise in accordance with the requirements of this specification.

3.2 Material. Unless otherwise specified herein, jaw cable grips shall be made of a carbon or alloy steel of commercial grade. All materials shall be of uniform quality and free of injurious defects. Strength and other physical properties shall be suitable for the purpose intended.

3.3 Fabrication. The cable grip shall be fabricated from separate parts which shall be forged or stamped at the manufacturer's option. The clearance between actuating members and their respective pivots, guides, pins, or holes over which they fit, shall be sufficiently close so as to preclude misalignment of the members, but not so close as to produce excessive friction or binding. Pins, when used as fulcrum or pivot points serving to join the movable members or linkage, shall be provided with a means for positive retention. When used, rivets shall be firmly set and neatly headed to preclude any loosening during operation. The component parts shall be of rugged construction to withstand continuous service operation under conditions of maximum safe loads (see tables I and II) for the specific size.

TABLE I. Type I, style A, parallel jaw grip, bronze-lined or steel; smooth jaws

| Size No. | Outside diameter of electrical wire, power cable, wire (plain) or wire rope | | Weight* (nominal) | Maximum safe load | Proof load |
|---|---|---|---|---|---|
| | Range | | | | |
| | Minimum | Maximum | | | |
| | Inches | Inches | Pounds | Pounds | Pounds |
| 4 | 0.081 | 0.200 | 1-1/2 | 1,500 | 1,725 |
| 6 | .102 | .373 | 3-1/4 | 4,500 | 5,175 |
| 8 | .156 | .312 | 1-3/4 | 5,000 | 5,750 |
| 9 | .162 | .552 | 6-1/4 | 8,000 | 9,200 |
| 12 | .250 | .500 | 3-5/16 | 7,500 | 8,625 |
| 13 | .312 | .625 | 18 | 15,000 | 17,250 |
| 14 | .375 | .750 | 4-7/8 | 10,000 | 11,500 |
| 16 | .500 | .750 | 18 | 15,000 | 17,250 |
| 17 | .562 | 1.062 | 9 | 12,500 | 14,375 |
| 18 | .625 | .875 | 7-3/4 | 8,000 | 9,200 |
| 19 | .750 | 1.000 | 18 | 15,000 | 17,250 |
| 24 | .625 | .625 | 27 | 20,000 | 23,000 |
| 25 | .750 | .750 | 27 | 20,000 | 23,000 |
| 26 | 1.000 | 1.000 | 27 | 20,000 | 23,000 |
| 27 | 1.250 | 1.250 | 27 | 20,000 | 23,000 |
| 28 | 1.500 | 1.500 | 32 | 25,000 | 28,750 |

* The maximum allowable weight shall not exceed the nominal weight by more than 10 percent.



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TABLE II. Type II, eccentric jaw

| Size No. | Outside diameter of electrical wire, power cable, wire (plain) or wire rope | | Weight[a] (nominal) | Maximum safe load | Proof load |
|---|---|---|---|---|---|
| | Range | | | | |
| | Minimum | Maximum | | | |
| | Inches | Inches | Pounds | Pounds | Pounds |
| 71 | 0.060 | 0.250 | 1 | 2,500 | 2,875 |
| 73 | .125 | .500 | 2-1/2 | 5,000 | 5,750 |
| [aa]75 | .250 | .750 | 3-1/2 | 8,000 | 9,200 |
| [aa]76 | .125 | .500 | 2-1/2 | 5,000 | 5,750 |

[a] The maximum allowable weight shall not exceed the nominal weight by more than 10 percent.
[aa] Furnished with safety latch as specified in 3.5.

3.4 Design. The cable grip shall consist essentially of a body piece, a pair of jaws, and a series of linkages, levers, and pins. A guide shall be provided which will limit the movement of the pulling shackle or clevis to the general direction of the longitudinal axis of the jaws. Outward motion of the pulling shackle or clevis shall cause the jaws to close. Additional outward pull on the pulling shackle or clevis shall cause the pressure between the jaws to increase. Conversely, inward motion of the pulling shackle or clevis shall cause the jaws to separate and open. In operation, the wire, cable, or wire rope is placed between the jaws in a longitudinal direction. The jaws shall be designed to firmly grip the wire, cable, or wire rope with a minimum of damage to the wire and with slippage of no more than 1/2 inch when pressure is applied. A pulling shackle or clevis shall be provided at the outer end of the actuating linkage which will accommodate a hoist or sling hook for use with a block and tackle.

3.5 Safety latch. When specified (see 6.1), a movable safety latch shall be provided on type II, sizes number 71 and 73 cable grips. The safety latches shall be so designed that when engaged, they entrap the wire, cable, or wire rope within the jaws of the cable grip. Suitable means shall be provided to retain the safety latch in the engaged position and prevent accidental release of the latch due to jarring, dropping, or other accidental blows. When released, the safety latch shall not obstruct the normal loading or unloading of the wire, cable, or wire rope.

3.6 Capacity. The jaws shall readily accept and firmly grip the range of sizes from the largest to the smallest size wire, cable, or wire rope listed in tables I and II for the size number of cable grip specified.

3.7 Proof load. The cable grip shall be subjected to the applicable proof loads in accordance with the test procedure in 4.4. Each cable grip shall be capable of transmitting to the wire, cable, or wire rope, the amount of proof load listed in tables I and II for the size number of cable grip specified. While the full proof load is applied, the various parts of the cable grip shall display no visible twisting, buckling, or other signs of physical weakness and shall grip the wire, cable, or wire rope with slippage of no more than 1/2 inch. After full proof load is released, the various parts of the cable grip shall work freely and display no distortion, bending, or other damage.

3.8 Plating. All parts, except the gripping surface of the movable jaw, shall be protected against rusting or corrosion by zinc or an electrogalvanized finish not less than 0.0003 inch in thickness.

3.9 Type I, parallel jaw. Type I cable grip shall be provided with two jaws which shall be aligned parallel to one another when closed on any size wire from the smallest to the largest diameter within the range specified.



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3.9.1 Type I, style A, smooth jaws. Type I, style A grips shall be furnished with either bronze-lined or steel jaws, having smooth gripping surfaces, and shall be capable of preventing slippage of wire, cable, or wire rope. When bronze-lined jaws are furnished, the gripping surfaces of the jaws shall have bronze liners permanently affixed. When steel jaws are furnished, the jaws shall be made entirely of steel. Gripping surfaces shall be oval, round, or V-shaped in cross section, except size 4 which shall have one flat and one V-shaped jaw. Style A cable grips shall be similar to figures 1 and 2 and shall conform to table I.

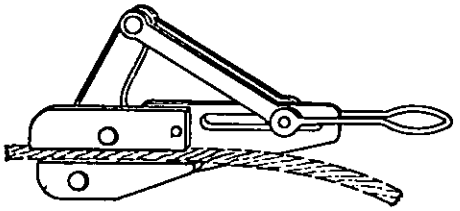


FIGURE 1.—Type I, parallel jaw.

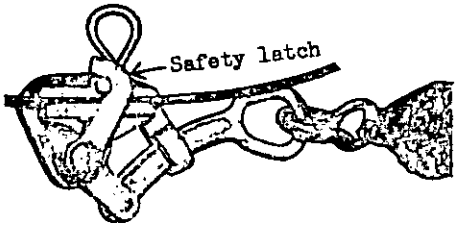



FIGURE 2.—Type I, parallel jaw with safety latch.

3.10 Type II, eccentric jaw. Type II cable grip shall be provided with two jaws for gripping the end of cables where the gripped section will be removed after tensioning. One jaw shall be fixed and an integral part of the body; the other shall be a movable jaw having an eccentric or cam-type action. These jaws shall be capable of closing and holding any size wire from the smallest to the largest diameter within the range specified. Type II cable grips shall be similar to figure 3 and shall conform to table II.

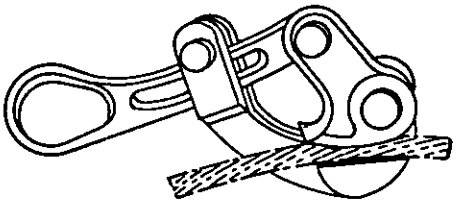


FIGURE 3.—Type II, eccentric jaw.



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3.11 Marking. Each cable grip shall be permanently marked in a plain and legible manner with the manufacturer's name or with a trademark of such known character that the source of manufacture may be readily determined. In addition, each grip shall be permanently marked with the applicable capacity (minimum and maximum) and the maximum safe load.

3.12 Workmanship. The workmanship shall be first class in every respect. On forged parts, all flashing shall be neatly removed and surfaces shall be free from scale, pits, nodules, burrs, fins, or projections. Sharp edges of stamped parts shall be neatly and smoothly rounded, and all burrs shall be removed. All parts shall be completely plated with the exception of the gripping surface of the movable jaw which shall not be plated. The grips shall operate freely and smoothly with a minimum of lost motion in the actuating linkage. The fulcrum pin shall be tightly fitted. All rivets shall be tight.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection of materials and components. In accordance with 4.1, the supplier is responsible for insuring that materials and components used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein, or, if none, in accordance with this specification.

4.2 Sampling procedures. Sampling procedures shall be in accordance with MIL-STD-105. Data for sampling shall be as stated in table III. All cable jaw grips of the same type and size offered for delivery at one time shall be considered a lot for purposes of inspection.

TABLE III. Sampling data

| Category | Sample unit | Inspection level | Acceptable quality level (AQL) | AQL expressed in terms of | Reference |
|---|---|---|---|---|---|
| Visual examination | 1 each | II | 4.0 | Defects per 100 units | 4.3.1 |
| Testing | 1 each | S-2 | 1.5 | Percent defective | 4.4.1 |
| Packaging and preservation | One container | S-2 | 4.0 | Defects per 100 units | 4.5 |

4.3 Examination.

4.3.1 Visual examination. Each sample unit shall be examined for any nonconformance in design, material, finish, construction, workmanship, and marking. If the number of defects in the sample exceeds the acceptance number specified in MIL-STD-105, the entire lot of grips shall be considered defective.

4.3.2 Dimensional examination. Each sample unit shall be examined for any nonconformance with dimensional requirements.



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4.4 Test.

4.4.1 Proof-load test. Grips shall be subjected to a tensile test to determine compliance with 3.6. The maximum speed of jaw separation of the tensile-testing device shall not exceed a rate of 1 foot per minute for the test. Grips shall be tested using the type wire, cable, or wire rope for which the particular grip under test was designed. In the event that the tensile strength of any specific type of wire, cable, or wire rope be less than the proof load specified for the cable grip being tested, a wire, cable, or steel rod which will withstand the proof load specified shall be substituted for test purposes.

4.4.1.1 During proof load. While proof load is applied to the maximum value indicated for the particular size cable grip under test, inspect to ascertain whether any parts display visible twist, buckle, or other sign of physical weakness. Inspect to determine slippage between the grip and the wire, cable, or wire rope used for test (see 3.7).

4.4.1.2 After release of proof load. Inspect to determine whether any part has become distorted, bent, or sustained other damage during test. Inspect the wire, cable, or wire rope used during test to ascertain whether cable grip caused damage which would be detrimental to the intended purpose of the wire, cable, or wire rope. The type II grip is exempt from this requirement.

4.4.1.3 Lot inspection. If any sample unit fails to meet any requirement for which tests are made, it shall be considered defective. If the number of defective sample units in the sample exceeds the acceptance number specified in MIL-STD-105, the entire lot of grips shall be considered as having failed to meet the requirements of the specification.

4.5 Inspection of preparation for delivery. Inspection shall be made to determine compliance with the requirements of PPP-P-40.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be in accordance with PPP-P-40. The level of preservation, packaging, and packing shall be level A, B, or C, as specified (see 6.1).

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and style of grip required (see 1.2).
(c) Size of grip required (see applicable table).
(d) Specify safety latch, if required (see 3.5).
(e) Level of preservation, packaging, and packing (see 5.1).

| MILITARY CUSTODIANS: | User activity: | Preparing activity: |
|---|---|---|
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